Computing Detention Pond Routing

Detention pond routing is the process of passing a flood hydrograph through a storage reservoir or detention pond. This process changes the pattern of flow with respect to time but conserves volume. The purpose of detention pond routing is usually to reduce the peak flow to a predetermined level, or to delay the peak. The routing procedure used by Hydraflow Hydrographs Extension is known as the Storage Indication method and begins with a stage-storage-discharge relationship, an inflow hydrograph, and the following equation:

$$I - O = \frac{ds}{dt}$$

Where:

I = inflow

O = outflow

ds/dt = change in storage

Hydraflow Hydrographs Extension first uses the specified stage-storage-discharge table to internally plot a curve of 2s/dt + O. It then computes the outflow hydrograph using a procedure similar to the following example.

The following table contains values for sample detention pond calculations.

Time (min) (1)	Ii (cfs) (2)	Ij (cfs) (3)	2s/dt-Oi (4)	2S/dt+Oj (cfs) (5)	Outflow (cfs) (6)
0	0	24	0	-	0
4	24	95	4	24	10
8	95	206	33	123	45
12	206 +	345 +	174 = 725	334	80
16	345	500	439	725	143
20	500	655	884	1284	200
24	655	794	1509	2039	265
28	794	905	2292	2958	333
32	905	976	3239	3991	376
36	976	1000	4310	5120	404

40	1000	976	5426	6286	430
44	976	905	6502	7402	450
48	905	848	7453	8383	465
52	848	736	8252	9206	477
56	736	638	8866	9836	485
60	638	554	9260	10240	490
64	554	480	9468	10452	492
68	480	417	9514	10502	494<
72	417	0	10411	491	

Routing procedure.

- Column 1 and column 2 are read from the inflow hydrograph.
- Column 3 is the inflow at time j.
- Column 4 is column 5 - 2 x column 6.
- Column 5 for j is [column 2 + column 3 + column 4]j.
- Column 6 is computed by straight-line interpolation from the plot of 2S/dt + O vs. O.